

Question 1

Explain the difference between EPC (4G) and 5GC (5G).

The EPC is a core network that is mainly used in 4G systems whereas the 5GC is a core network designed for 5G. EPC was mainly built to support mobile services off broadband while 5GC is flexible and used in more advanced features.

Question 2

Why is network slicing important in 5G? Provide real-world examples.

Network slicing is important because it allows a single 5G network to be divided into multiple virtual networks. This allows 5G to be more efficient and receive the necessary resources they need. Hospitals use network slices for surgeries that require low latency, while cities use another slice on traffic sensors, etc.

Question 3

What is Multi-access Edge Computing (MEC), and how does it reduce latency?

MEC is distributed computing that brings cloud services and processing power to the edge of a network. When processing information closer to where it is generated, This can reduce the time it will take for the data to travel which results in reduced latency.

Question 4

Describe the role of AI in dynamic resource allocation.

Artificial intelligence helps communication networks allocate resources by analyzing traffic patterns and adjusting bandwidth in real time. Improving network performance and reduces delays without needing a person intervening.

Question 5

In SBA, what is the role of the Network Repository Function (NRF)?

In SBA, the network repository function serves as a directory for networking functions. They are allowed to keep track of available network services and use this to discover or even communicate with each other. It also makes the 5G core network efficiently and more flexible.

